Problem-Solution Fit

Date	5 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:
Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS]	**6. CUSTOMER LIMITATIONS** EG. BUDGET, DEVICES **[CL]**	**5. AVAILABLE SOLUTIONS** PROS & CONS **[AS]**
i. People residing in flood vulnerable regions. ii. Emergency and Disaster Response Teams. iii. Landowners and Farmers. iv. Municipal and Government Authorities. v. Climate and Environmental Analysts.	• Unstable internet connectivity in remote locations. • Limited awareness of advanced digital forecasting tools. • Inability to afford premium flood monitoring systems. • Heavy reliance on publicly available weather reports and alerts.	**Current Solutions** • Weather department flood advisories. • Conventional weather observation methods. • Television, radio and online news updates. **Advantages** • Free access to the public. • Information provided by trusted government agencies. **Disadvantages** • Delayed updates • Predictions are often broad and not location-specific.

- Limited use of machine learning for accurate flood forecasting.

2. PROBLEMS / PAINS
+ ITS FREQUENCY **[PR]**

- Difficulty in identifying flood risks before they occur.
- Sudden heavy rainfall often leads to unexpected flooding.
- Insufficient access to timely and reliable flood warning systems.
- Most common during monsoon seasons and periods of intense rainfall.

9. PROBLEM ROOT / CAUSE [RC]

- Highly unpredictable weather patterns.
- Intense monsoon rainfall and changing climate conditions.
- Increasing impact of climate change on rainfall intensity.
- Limited availability of accurate, real time flood prediction technologies.

7. BEHAVIOR
+ ITS INTENSITY **[BE]**

- Frequently monitors weather forecasts and rainfall reports during the rainy season.
- Follows safety guidelines and prepares emergency measures whenever flood alerts are issued.
- Actively tracks flood conditions when heavy rainfall is expected.

3. TRIGGERS TO ACT [TR]

i. Forecasts indicating intense or prolonged rainfall.

ii. Increasing water levels in rivers, lakes or reservoirs.

iii. Unexpected changes in weather conditions.

iv. The need to receive early flood warnings and prepare in advance.

4. EMOTIONS
BEFORE / AFTER **[EM]**

Before

- Concerned about possible flooding.
- Unsure of the potential flood risk.
- Stressed due to

10. YOUR SOLUTION [SL]

Rising Waters is a Machine Learning powered flood prediction system that evaluates weather-related parameters such as annual rainfall, cloud visibility, seasonal rainfall, average rainfall and other meteorological data using advanced classification algorithms including Decision Tree, Random Forest and KNN. The best performing model is integrated into a Flask web application, providing instant flood risk predictions as Flood Risk or No Flood Risk. This enables authorities, emergency

8. CHANNELS OF BEHAVIOR [CH]

ONLINE
i. Rising Waters Flood Prediction Web Portal.
ii. Online weather forecasting platforms.
iii. Official disaster management and government websites.

OFFLINE

- Local television weather bulletins.
- Radio weather and emergency broadcasts.
- Public announcements issued by disaster management authorities.

unpredictable weather conditions. **After** • Better informed through timely flood predictions. • Ready to take preventive actions. • Reassured and confident in making safety decisions.	response teams and residents to make informed decisions, improve disaster preparedness and reduce the impact of flooding.	